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ostartsection section1@-3.5ex oplus -1ex ominus -.2ex2.3ex oplus.2ex*Mass–Hierarchy Derivation in the SAT.O.
ostartsectionsubsection2@-3.25exoplus -1ex ominus -.2ex1.5ex oplus .2ex*1. Bare filament mass scale m_0
In the SAT.O–Blockwave model the fundamental filament tension is denoted by T (units: energy per length,

since $T \ell_f$ has units of energy (ML^2T^{-2}), dividing by c^2 yields a mass (M). Thus

ostartsectionsubsection2@-3.25exoplus -1ex ominus -.2ex1.5ex oplus .2ex*2. Topological invariant Q
Define the topological invariant

a dimensionless integer (or near-integer) classifying filament bundle topology. For simplicity one may approximate

for minimal excitations.

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The dominant (“topological”) contribution to the effective mass is postulated as

This expresses the idea that the bare filament mass is suppressed by the inverse of topological complexity.
ostartsectionsubsection2@-3.25exoplus -1ex ominus -.2ex1.5ex oplus .2ex*4. Induced inertial mass m_{ind}
Coupling of the filament bundle to the timesheet (foliation) field $u^\mu(x)$ yields an induced inertia. In leading order

where:

- g_1 is a dimensionless drag/inertia coupling constant,
- M^2 is the mass-gap squared of the u^μ fluctuations: $\omega^2 = k^2 + M^2$,
- E_{overlap} is the energy scale at which the filament bundle overlaps / sources the u^μ field,
- C is a dimensionless geometric constant.

Under units: $[E_{\text{overlap}}] = \text{energy} = ML^2T^{-2}$; dividing by M^2 (units T^{-2}) yields $ML^2T^{-2} \times T^2 = ML^2$; multi
ostartsectionsubsection2@-3.25exoplus -1ex ominus -.2ex1.5ex oplus .2ex*5. Total effective mass

ostartsectionsubsection2@-3.25exoplus -1ex ominus -.2ex1.5ex oplus .2ex*6. Topological dominance condition
We demand:

That is,

Given numerical values for g_1 , E_{overlap} , M^2 , m_0 , Q one can bound the regime of validity.

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Particle excitations labelled by a topological Q-charge Q and a vibrational quantum number N obey:

where k, β, a are calibration constants (with units M^2). In SAT variables we set

for a reference topological charge Q_0 , and solve:

Matching ground-state masses fixes k, β, a ; excited states $N = 1, 2, \dots$ yield the mass spectrum hierarchy.

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From the $\mathcal{L}_{\theta_4}$ block we define the canonical scalar field $\phi \equiv \sqrt{f_\theta} \theta_4$. Its mass squared is: